

Esmaeil Mousavi

Salt Lake City, UT — esmaeilmousavi@weber.edu — linkedin.com/in/mousavi-ai — 775-300-9794

Professional Experience

Weber State University – Ogden, UT

Researcher – AI/ML & Backend Systems

May 2025 – Present

- Leading faculty-mentored applied research through the *navNote* platform, an AI-native system for enterprise field operations.
- Research in large-scale image understanding, predictive analytics, autonomous task orchestration, and scalable backend architectures.
- Faculty-mentored research transitioned into an AI-native startup, **navNote LLC** (navNote.net), co-founded with an IBM engineer (U.S. citizen); Esmaeil is the primary/major equity holder. Headquartered in New York City with a Utah-based tech office. Established partnerships (Kroger, mimik Technologies, Perplexity, iHub Utah), secured \$10K+ in state awards, formed an executive board with leaders from multi-billion-dollar companies, and preparing enterprise retail deployments.

Assistant Researcher – AI in Autonomous Vehicles, Systems & Technologies

Oct 2022 – May 2025

- Conducted applied research in autonomous navigation and perception using reinforcement learning and 3D LiDAR.
- Improved obstacle recognition accuracy by 25% via SWARM learning models; organized 25+ research tours and lectures.

Graduate Assistant Teacher – CS 6300 (Route Planning & Navigation)

Mar 2023 – Dec 2023

- Co-designed and supported a graduate-level course; led hands-on labs using TurtleBots and F1-Tenth platforms.

Assistant Researcher – ML in Acoustic Denial & Cybersecurity

Jan 2023 – May 2025

- Simulated 100+ acoustic denial-of-service attacks on HDDs; reduced failure rates by 40% using ML-based mitigation.
- Secured competitive funding to present findings at NCUR 2025.

National Center for Water Quality Research – Tiffin, OH

Assistant Researcher – ML in Predictive Modeling

Nov 2021 – Sep 2022

- Applied ML to predict Sandusky River water quality trends up to eight years ahead using decade-long datasets.

Data Science & Visualization Specialist

Dec 2021 – Sep 2022

- Processed and visualized large-scale environmental sensor data for state and regional monitoring.

Education

Weber State University – Ogden, UT *Bachelor of Science in Computer Science* Expected Graduation: Dec 2026

Academic Distinctions & Honors

- **7th Place Nationally**, International Mathematics Competition (IMC Olympiad).
- **Top-Ranked** in the Entrance Examination for the National Organization for Development of Exceptional Talents (NODET).

Publications, Patents & Selected Projects

- *Machine Intelligence Solution for Acoustic Denial of Service Attacks on HDDs* (WSU — NCUR 2025 Submission).
- *Forecast on Sandusky River's Particulate Phosphorus Ratio* (NCWQR Collaboration, 2022).
- *Two-Stroke X-Shaped Engine* (WIPO #WO/2020/026037).
- Designed and deployed a real-time autonomous vehicle obstacle detection system; organized applied AI/AV research tours.

Leadership, Service & Professional Affiliations

- **Ambassador, IBM Z Mainframe Systems** (Oct 2021 – Present): Led student workshops and enterprise systems briefings.
- **Senator, College of Engineering, Applied Science & Technology, Weber State University** (Mar 2023 – May 2024): Represented engineering students; advised on academic resources and IT policy.
- **Member, Faculty Senate Committees, Weber State University** (Oct 2023 – May 2024): Served on Academic Resources and IT Advisory committees.
- **Student Advisor, NVIDIA & HPE Swarm Learning Project** (Jul 2023): Guided collaborative model optimization strategies.
- **Organizer, AI & Autonomous Vehicles Research Tours, Weber State University** (Nov 2023).
- **Member, Harvard SEAS Computer Science Colloquium** (Oct 2021 – Present); **Publisher, MIT Open Learning** (open.mit.edu).
- **Member, Association for Computing Machinery (ACM®).**